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MAJOR PROJECT PHASE-I REPORT

ON

"Patch-Based Unimodal Cancer Anomaly Detection in tissue Images Using Deep Learning Models"

Submitted in the partial fulfilment of requirements for the award of Degree

B.E in Computer Science and Engineering

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2.	Creating collaborative learning environment that ignites the critical thinking in students and leading to the innovation.
3.	Establishing Industry Institute relationship to bridge skill gap and make them industry ready and relevant.
4.	Mentoring students to be socially responsible by inculcating ethical and moral values.

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ABSTRACT

The proposed research focuses on patch-based cancer detection in histopathology images using deep learning techniques. Whole Slide Images (WSI) are large and complex, making direct processing computationally expensive. To address this, the system extracts smaller patches from WSI and utilizes convolutional neural networks such as ResNet, DenseNet, and EfficientNet to classify them as tumor or normal. A weakly supervised approach is adopted by using slide-level labels instead of detailed annotations, significantly reducing the dependency on expensive manual labeling. To improve performance, techniques such as data augmentation, context-aware patch extraction, and anomaly detection are incorporated to identify deviations from normal tissue patterns. Additionally, Explainable AI using Grad-CAM generates heatmaps highlighting important regions to ensure clinical interpretability.

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CHAPTER 1

INTRODUCTION

A patch-based unimodal deep learning anomaly detection framework is proposed for automated, explainable breast cancer screening using histopathology whole-slide images (WSIs), shifting the traditional paradigm toward unsupervised outlier detection to eliminate the dependency on extensive malignant datasets. The framework partitions tissue images into structured, overlapping multi-scale patches to capture localized cellular morphology alongside global spatial architecture. Feature extraction is conducted via parallel deep learning backbones, leveraging an EfficientNet architecture to identify cell-level density and a ResNet-50 model with residual skip connections to preserve ductal boundaries. To resolve the clinical transparency gap, an explainable artificial intelligence (XAI) block utilizing Gradient-weighted Class Activation Mapping (Grad-CAM) generates real-time visual attention heatmaps over suspicious features. Ultimately, the system evaluates a unified anomaly score against a healthy tissue manifold to flag critical structural and cellular abnormalities—including nuclei pleomorphism, tumor cell density clusters, ductal structure distortions, and necrotic dead zones—providing a highly robust, scalable, and interpretable tool for routine digital pathology workflows.

1.1 Introduction to Breast Cancer and Histopathology

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. It occurs due to the uncontrolled growth of abnormal cells in breast tissues, which may spread to other parts of the body if not detected early. Early diagnosis plays an important role in improving survival rates and treatment outcomes [5], [12].

Histopathological examination is considered the gold standard for breast cancer diagnosis. In this process, tissue samples obtained through biopsy are examined under a microscope to identify abnormalities in tissue structures and cellular morphology. Although highly reliable, manual analysis is time-consuming, labor-intensive, and dependent on the expertise of pathologists. Inter-observer variability may also lead to inconsistencies in diagnosis [17], [18].

Recent advancements in Artificial Intelligence (AI) and deep learning have enabled the development of automated systems that assist pathologists in detecting cancerous tissues accurately and efficiently.

1.2 Growth of Digital Pathology and Artificial Intelligence in Healthcare

Digital pathology and Whole Slide Imaging (WSI) technologies allow histopathological slides to be converted into high-resolution digital images for computational analysis. These digital slides can be analyzed using deep learning techniques for automated cancer detection.

Convolutional Neural Networks (CNNs) have shown excellent performance in medical image analysis due to their ability to automatically learn important image features. Advanced architectures such as ResNet, DenseNet, EfficientNet, and VGG19 are widely used for breast cancer classification and histopathological image analysis [1], [4], [6], [13].

AI-assisted diagnostic systems help improve diagnostic accuracy, reduce workload, and support pathologists in clinical decision-making [5], [14].

1.3 Challenges in Traditional Histopathological Image Analysis

(A) Large Size of Whole Slide Images

Whole Slide Images are extremely large and require high computational resources for processing. To overcome this issue, researchers commonly use patch-based approaches where large images are divided into smaller patches for efficient analysis [9], [20].

However, analyzing patches independently may lead to loss of global contextual information, which can affect classification performance.

(B) Dependency on Large Labeled Datasets

Most supervised deep learning models require a large amount of labeled training data to achieve high accuracy. In histopathological image analysis, obtaining detailed annotations requires expert pathologists who manually identify cancerous regions within tissue slides. This annotation process is expensive, time-consuming, and difficult to scale for large datasets [19], [23].

The scarcity of accurately labeled medical datasets significantly limits the development of robust and generalized AI systems for healthcare applications.

(C) Lack of Interpretability

Deep learning models often function as "black-box" systems that provide predictions without clearly explaining their decisions. In healthcare applications, interpretability is important for gaining clinician trust and ensuring reliable diagnosis [17], [18].

1.4 Patch-Based Deep Learning for Histopathological Analysis

Patch-based learning is one of the most effective approaches for analyzing histopathological images. In this method, Whole Slide Images are divided into smaller fixed-size patches, which are processed individually using CNN models for feature extraction and classification [8], [9].

This approach reduces computational complexity and enables efficient analysis of large histopathological images while allowing models to focus on important local tissue features relevant to cancer diagnosis.

Several studies have shown that patch-based deep learning models achieve high accuracy in breast cancer classification tasks. These methods also improve scalability and allow efficient training of deep learning architectures on large medical imaging datasets [4], [7].

However, since patches are analyzed independently, some global contextual information from the original slide may be lost, which can affect overall classification performance [20].

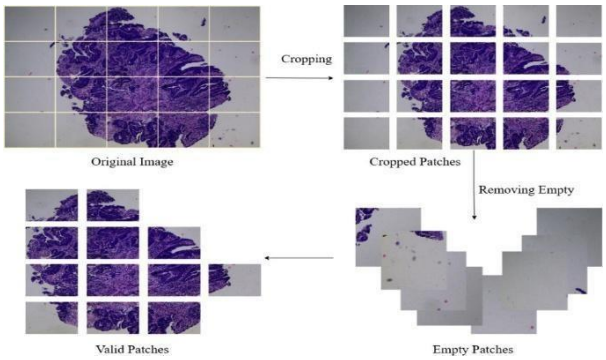


Fig 1.1: Patch-Based Histopathological Image Processing

1.5 Weakly Supervised Learning and Anomaly Detection

To overcome the limitations associated with large-scale medical annotations, researchers have

increasingly focused on weakly supervised learning and anomaly detection techniques.

Weakly supervised learning reduces annotation dependency by using slide-level labels instead of detailed patch-level labels. This significantly decreases manual labeling effort while still enabling effective model training [3], [10].

Anomaly detection techniques have also gained significant attention in medical imaging applications. Instead of learning from both normal and abnormal samples, anomaly detection models primarily learn the characteristics of normal tissue patterns and identify deviations as potential abnormalities or cancerous regions [3], [11].

These methods are particularly useful for detecting rare or previously unseen cancer patterns that may not be well represented in training datasets. Anomaly detection frameworks also improve scalability and reduce dependency on extensive labeled datasets, making them suitable for real-world healthcare applications.

The proposed project focuses on developing a patch-based unimodal anomaly detection framework for breast cancer detection using histopathological images. The system aims to improve diagnostic accuracy while minimizing annotation requirements and computational complexity.

1.6 Targeted Structural and Cellular Tissue Anomalies

To build a robust unimodal anomaly detection system that trains primarily on the manifold of healthy tissue architecture, the framework must precisely isolate and flag deviations across four critical pathological markers. These abnormalities represent the fundamental cellular and structural distortions introduced during the progression of breast carcinomas:

(A) Nuclei Pleomorphism (Nuclear Atypia)

In normal breast tissue, epithelial cell nuclei display uniform size, smooth oval contours, and regular chromatin distribution. Nuclei pleomorphism represents a profound deviation where malignant cells show marked variability in size, shape, and staining intensity. Under high magnification, pleomorphic nuclei exhibit macrokaryosis (severe enlargement), hyperchromasia (darker staining due to dense DNA content), and irregular, angular borders. The proposed system utilizes fine-grained feature mapping to capture variations in nuclear eccentricity and circularity

to automatically flag these atypical groupings as localized anomalies.

(B) Ductal Structure Distortion

Healthy mammary glands are characterized by well-defined, organized ductal loops lined by an orderly bilayer of epithelial and myoepithelial cells. During the onset of Ductal Carcinoma In Situ (DCIS) and its transition into Invasive Ductal Carcinoma (IDC), these boundaries undergo severe distortion. Malignant cells proliferate uncontrollably, filling the central lumen, breaking through the protective basement membrane, and infiltrating the surrounding connective stroma. This results in a complete loss of macro-level tissue geometry. The model's multi-scale patch extraction ensures that these large-scale architectural breakdowns are captured at low magnifications.

(C) Tumor Cell Density Clusters

A signature manifestation of breast malignancy—particularly in IDC patches—is the rapid formation of high-density cellular clusters. Unlike the sparse, fat-dominated spatial layout of healthy connective stroma, cancerous regions feature a highly dense packing of overlapping cell bodies. This high concentration of nuclei creates a chaotic texture profile in the image. By evaluating the relative ratio of cellular material to surrounding stroma across overlapping patches, the anomaly detection framework triggers elevated deviation scores in regions showing sudden cellular overcrowding.

(D) Necrosis

In highly aggressive, rapidly growing breast tumors, the metabolic demands of the dense cellular clusters often outpace the local blood supply. This oxygen and nutrient deprivation triggers comedonecrosis—areas of dead tissue typically located within the center of distorted malignant ducts. On H&E-stained slides, necrotic zones appear as granular, structural debris lacking viable cell nuclei or distinct boundaries. Traditional classification models frequently misidentify these zones due to the absence of active cell shapes; however, the proposed unsupervised framework identifies this "textural chaos" and lack of normal cellular structure as a prominent out-of-distribution anomaly.

1.7 Deep Learning Architectures Used in the Proposed System

The proposed system utilizes advanced Convolutional Neural Network architectures such as ResNet, DenseNet, and EfficientNet for feature extraction and cancer classification.

(A) ResNet

Residual Networks (ResNet) introduce skip connections that allow gradients to flow directly across layers during training. This helps overcome the vanishing gradient problem and enables the training of very deep neural networks with improved classification accuracy [1], [13].

(B) DenseNet

DenseNet establishes dense connectivity between layers, enabling feature reuse and improving information propagation throughout the network. This architecture reduces redundancy and improves learning efficiency.

(C) EfficientNet

EfficientNet uses compound scaling techniques to optimize network depth, width, and image resolution simultaneously. It provides an effective balance between computational efficiency and classification accuracy, making it highly suitable for large-scale medical image analysis tasks [1], [6].

(D) VGG19

VGG19 is another widely used deep learning architecture in histopathological image classification. It uses multiple stacked convolutional layers for extracting complex visual features from tissue images and has shown promising results in breast cancer diagnosis applications [4].

These architectures have demonstrated strong performance in various medical imaging tasks due to their ability to learn complex pathological patterns effectively.

1.8 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) has become increasingly important in AI-driven healthcare systems because it improves transparency, reliability, and trustworthiness of automated predictions.

The proposed system integrates Gradient-weighted Class Activation Mapping (Grad-CAM), an Explainable AI technique that generates visual heatmaps highlighting important image regions

responsible for model predictions. Grad-CAM enables clinicians to verify whether the model focuses on clinically relevant pathological structures while making diagnostic decisions [17], [18].

Other explainability techniques such as saliency maps, feature visualization, and attention mechanisms are also widely explored in medical image analysis applications. These methods help bridge the gap between automated AI predictions and clinical interpretation. The integration of XAI techniques is essential for achieving clinical acceptance of AI systems because healthcare professionals require transparency and accountability before relying on automated diagnostic tools in real-world environments.

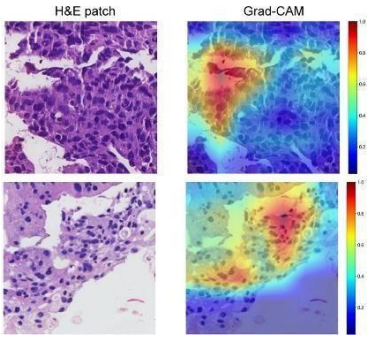


Fig 1.2: Grad-CAM Based Explainability in Cancer Detection

1.9 Overview of the Proposed System

The proposed project focuses on developing a patch-based unimodal anomaly detection framework for breast cancer detection using histopathological images.

The system combines patch-based image analysis, weakly supervised learning, deep learning architectures, and Explainable AI techniques to create an efficient and interpretable diagnostic framework. Preprocessing methods such as normalization and data augmentation are also incorporated to improve model robustness and generalization [21].

The proposed framework aims to improve diagnostic accuracy, reduce dependency on extensive annotations, and assist pathologists in efficient cancer detection.

CHAPTER 2

LITERATURE SURVEY

2.1 REVIEW OF EXISTING WORK

[1] **Sania Eskandari, Ali Eslamian, Nusrat Munia, Amjad Alqarni, and Qiang Cheng [2025] – "Evaluating Deep Learning Models for Breast Cancer Classification: A Comparative Study"** evaluated multiple deep learning architectures including CNNs and Vision Transformers for breast cancer classification. The study compared model performance based on accuracy, sensitivity, and computational efficiency using histopathological image datasets. The authors highlighted the importance of preprocessing techniques and model selection in improving diagnostic performance. However, the study faced challenges related to dataset imbalance and lack of interpretability in model predictions.

[2] **Fajar Walayat, Allah Ditta, Zafar Iqbal Karmani, Bilal Shoaib, Tehseen Mazhar, Muhammad Adnan Khan, and Rashid A. Saeed [2025] – "Deep Learning-Based Analysis of Mammographic Images for Breast Cancer Detection Using Transfer Learning"** proposed a transfer learning-based framework for breast cancer detection using mammographic images. The study utilized pre-trained CNN models along with noise filtering techniques and Grad-CAM heatmaps for explainability. Experimental results showed improved classification accuracy and reduced training time. However, the model was evaluated only on a local hospital dataset and showed performance degradation under noisy image conditions.

[3] **Jonas Dippel, Niklas Prenißl, Julius Hense, and Philipp Liznerski [2024] – "AI-based Anomaly Detection for Clinical-Grade Histopathological Diagnostics"** introduced an anomaly detection framework for histopathological image analysis using unsupervised learning techniques. The model was trained primarily on normal tissue samples and identified abnormal patterns as anomalies. The proposed approach improved robustness and reduced dependency on labeled datasets. However, the system could detect abnormalities without providing specific disease classification.

[4] **Anitha Ponraj, Palanigurupackiam Nagaraj, and Duraisamy Balakrishnan [2024] – "A Multi-Patch-Based Deep Learning Model with VGG19 for Breast Cancer Classifications in the Pathology Images"** proposed a multi-patch deep learning framework using the VGG19 architecture for breast cancer classification. Multiple image patches were analyzed to capture local and contextual tissue features effectively. The model achieved improved classification accuracy compared to single-patch approaches. However, the system performance varied across different magnification levels and datasets.

[5] **Istiak Ahmad and Fahad Alqurashi [2024] – "Early Cancer Detection Using Deep Learning and Medical Imaging: A Survey"** presented a comprehensive survey of deep learning techniques used for early cancer detection across multiple cancer types. The study reviewed CNNs, transfer learning, and segmentation models used in medical image analysis. The authors emphasized the importance of AI-based early diagnosis systems for improving patient outcomes. However, most existing systems lacked explainability and were specialized for individual cancer types.

[6] **Aadhi Aadhavan Balasubramanian, Salah Al-Heejawi, et al. [2024] – "Ensemble Deep Learning-Based Image Classification for Breast Cancer Subtype and Invasiveness Diagnosis"** proposed an ensemble learning framework combining multiple deep learning models such as VGG16 and ResNet50 for breast cancer subtype classification. The ensemble approach improved prediction robustness and reduced model bias. Experimental results demonstrated high classification accuracy and reliable subtype detection. However, the framework lacked integration into practical clinical workflows.

[7] **Ardavan Modarres, Erfan Ebrahim Esfahani, and Mahsa Bahrami [2023] – "GROUP MIXER: Patch-Based Group Convolutional Neural Network for Breast Cancer Detection"** introduced a lightweight patch-based CNN architecture using group convolution and channel shuffling techniques. The proposed GROUP MIXER model reduced computational complexity while maintaining high classification performance. The model effectively captured spatial relationships between tissue patches. However, slight reductions in overall accuracy were observed due to model compression.

[8] **Sergio Ortiz, Ignacio Rojas-Valenzuela, Fernando Rojas, et al. [2023] – "Novel Methodology for Detecting and Localizing Cancer Area in Histopathological Images Based on Overlapping Patches"** proposed an overlapping patch extraction technique combined with image processing methods for cancer localization. The approach improved localization accuracy by preserving contextual information between neighboring patches. Experimental results showed strong performance in identifying tumor regions. However, the model was sensitive to threshold tuning and was evaluated on limited datasets.

[9] **Aashita Rajput, Muskan Yadav, Sachin Yadav, et al. [2023] – "Patch-Based Breast Cancer Histopathological Image Classification Using Deep Learning"** developed a patch-based deep learning model for classifying histopathological biopsy images into healthy and cancerous categories. The approach focused on extracting local features from tissue patches using CNN architectures. The model achieved good classification accuracy while reducing noise from irrelevant image regions. However, the architecture struggled to capture complex biological variations in tissue structures.

[10] **Yinsheng He and Xingyu Li [2023] – "Whole-slide-imaging Cancer Metastases Detection and Localization with Limited Tumorous Data"** proposed a deep learning approach for metastasis detection using Whole Slide Images with limited labeled tumor samples. The study utilized reverse knowledge distillation techniques to improve learning efficiency and reduce dependency on large annotated datasets. The model achieved reliable detection accuracy while handling high-resolution images effectively. However, a small amount of tumor data was still required for training.

[11] **Ahad Alloqmani, Yoosef B. Abushark, and Asif Irshad Khan [2023] – "Anomaly Detection of Breast Cancer Using Deep Learning"** focused on anomaly detection using MobileNetV2 for identifying abnormal breast tissue patterns. The model learned normal tissue representations and detected deviations as cancerous abnormalities. The approach reduced dependency on labeled data and improved rare case detection. However, model performance relied heavily on publicly available datasets, which may not fully represent real-world clinical conditions.

[12] **Mai Alsammak and Marwa Khattabi [2023] – "Breast Cancer Screening: A Literature Review"** reviewed traditional and AI-based breast cancer screening methods including mammography, MRI, and deep learning techniques. The study discussed the role of early detection in reducing mortality rates and improving treatment outcomes. It also highlighted the challenges associated with false positives and limited access to automated screening systems in low-resource regions.

[13] **Xiaoyan Jiang, Zuojin Hu, Shuihua Wang, and Yudong Zhang [2023] – "Deep Learning for Medical Image-Based Cancer Diagnosis"** explored various deep learning architectures used for cancer diagnosis through medical imaging. The study reviewed CNNs, transfer learning, and hybrid models for classification and tumor detection tasks. It emphasized the importance of preprocessing and feature extraction in improving diagnostic performance. Challenges such as overfitting, limited annotated datasets, and lack of explainability were also discussed. The authors highlighted the need for more robust and interpretable AI systems for clinical applications.

[14] **Hirald Dwaraka Praveena, Shaik Saba Samreen, et al. [2023] – "Detection of Breast Cancer Using Deep Learning Techniques"** developed a CNN-based framework for detecting invasive ductal carcinoma regions in breast histopathological images. The study utilized deep learning models to automatically extract important tissue features and classify cancerous regions from biopsy images. Experimental results showed promising classification accuracy on benchmark datasets while reducing manual diagnostic effort. However, the model performance varied across different magnification levels, affecting consistency in cancer detection.

[15] **Salman Zakareya, Habib Izadkhah, and Jaber Karimpour [2023] – "A New Deep-Learning-Based Model for Breast Cancer Diagnosis from Medical Images"** proposed a hybrid deep learning architecture integrating feature extraction and attention mechanisms for breast cancer diagnosis. The model improved classification accuracy and computational efficiency. However, the performance was limited by the availability of high-quality annotated medical datasets.

[16] **Monika Subramanian, Nagarajan Ganesan, and SathishKumar Balasubramaniyan [2023] – "An Extensive Survey on Various Tumor Detection in Histopathological Images Using Deep Learning Techniques"** presented a survey of tumor detection techniques used in histopathological image analysis. The study reviewed various deep learning architectures,

segmentation methods, and preprocessing techniques used for tumor identification. It also highlighted computational challenges associated with processing high-resolution Whole Slide Images.

[17] **Yawen Wu, Michael Cheng, Shuo Huang, et al. [2022] – "Recent Advances of Deep Learning for Computational Histopathology: Principles and Applications"** reviewed recent developments in computational histopathology using deep learning. The paper discussed techniques for tissue segmentation, classification, color normalization, and prognosis prediction. Challenges such as staining variability, annotation dependency, and interpretability were also highlighted.

[18] **Alhassan Ali Ahmed, Mohamed Abouzid, and Elżbieta Kaczmarek [2022] – "Deep Learning Approaches in Histopathology"** reviewed the applications of deep learning techniques in automated histopathological diagnosis and cancer detection. The study discussed the use of CNNs, transfer learning, and segmentation models for analyzing tissue images and identifying cancerous regions. It highlighted the growing role of AI in improving diagnostic accuracy and reducing manual workload for pathologists. The paper also emphasized challenges such as clinical validation, limited annotated datasets, lack of interpretability, and the acceptance of AI systems in real-world healthcare environments.

[19] **Murtaza Ashraf, Willmer Rafell Quiñones Robles, Mujin Kim, and Mun Yong Yi [2022] – "A Loss-Based Patch Label Denoising Method for Improving Whole-Slide Image Analysis Using a Convolutional Neural Network"** proposed a loss-based patch denoising method called "LossDiff" to identify and remove mislabeled image patches during training. The method improved model generalization and classification accuracy by reducing the impact of noisy annotations. However, threshold tuning remained a challenge.

[20] **Ozan Ciga, Tony Xu, Sharon Nofech-Mozes, Shawna Noy, and Anne L. Martel [2021] – "Overcoming the Limitations of Patch-Based Learning to Detect Cancer in Whole Slide Images"** addressed the limitations of patch-based learning by incorporating spatial relationships between patches. The proposed method combined local and global contextual features for improved slide-level analysis and cancer detection. Experimental results showed enhanced classification performance and reduced false positives.

[21] **Massimo Salvi, Filippo Molinari, U. Rajendra Acharya, and Kristen M. Meiburger**

[2021] – "Impact of Stain Normalization and Patch Selection on the Performance of Convolutional Neural Networks" investigated the effects of stain normalization and patch selection techniques on CNN performance. The study demonstrated that proper preprocessing significantly improves classification accuracy and model robustness in histopathological image analysis.

[22] **Irum Hirra, Mubashir Ahmad, Iftikhar Ahmed Saeed, and Ahmed M. Alghamdi [2021] – "Breast Cancer Classification From Histopathological Images Using Patch-Based Deep Learning Modeling"** proposed a patch-based deep learning framework using CNNs and Deep Belief Networks for breast cancer classification. The approach improved computational efficiency and classification accuracy across different magnification levels. However, the model required high GPU resources for training.

[23] **Yasin Yari, Thuy V. Nguyen, and Hieu T. Nguyen [2020] – "Deep Learning Applied for Histological Diagnosis of Breast Cancer"** explored deep learning models for automated histological diagnosis of breast cancer using tissue images. The study evaluated various CNN architectures and preprocessing techniques to improve classification accuracy and feature extraction. The proposed approaches helped identify cancerous tissue patterns while reducing manual diagnostic effort. The study also highlighted the growing importance of AI-assisted diagnostic systems in healthcare.

[24] **Yuqian Li, Junmin Wu, and Qisong Wu [2019] – "Classification of Breast Cancer Histology Images Using Multi-Size and Discriminative Patches Based on Deep Learning"** proposed a multi-scale patch extraction framework for breast cancer histology image classification. The method used image patches of varying sizes to capture both cellular and tissue-level features. CNN models were used to learn pathological patterns for classifying benign and malignant tissues. Experimental results showed improved classification performance, but handling multiple patch sizes increased preprocessing complexity.

[25] **Md Zahangir Alom, Chris Yakopcic, et al. [2019] – "Breast Cancer Classification from Histopathological Images with Inception Recurrent Residual Convolutional Neural Network"** introduced a hybrid architecture combining Inception modules, recurrent layers, and residual connections for histopathological image classification. The model effectively captured spatial dependencies and improved classification accuracy. However, the architecture required large datasets and long training times due to its complexity.

2.2 LITERATURE REVIEW

SL NO	Author(s)	Year	Title	Methodology	Gaps
1	Sania Eskandari, Ali Eslamian, Nusrat Munia, Amjad Alqarni, Qiang Cheng	2025	Evaluating Deep Learning Models for Breast Cancer Classification: A Comparative Study	Compares eight deep learning models including standard CNNs and Vision Transformers (ViTs). **Dataset used:** Breast Histopathology Images (focussing on IDC): 277,524 patches extracted from 162 Whole Slide Images (WSI).	Suffers from dataset class imbalance. Clinicians still struggle to interpret model decisions transparently.
2	Fajar Walayat, Allah Ditta, Zafar Iqbal Karmani, Bilal Shoaib, Tehseen Mazhar, Muhammad Adnan Khan, Rashid A. Saeed	2025	Deep Learning-Based Analysis of Mammographic Images for Breast Cancer Detection Using Transfer Learning	Uses five pre-trained CNN models with noise filters. Adds Grad-CAM heatmaps for explainability. **Dataset used:** Private Local Dataset of 500 images collected from a local hospital in Pakistan.	Tested on a single local hospital dataset. Accuracy drops under noise or image rotations.
3	Jonas Dippel, Niklas Prenißl, et al.	2024	AI-based Anomaly Detection for Clinical-Grade Histopathological Diagnostics	Uses unsupervised anomaly detection trained only on normal tissue images. **Dataset used:** Charité Biopsy Datasets of 17 million images from 5,423 cases (Stomach and Colon cohorts).	Flags unknown cases as anomalies well. Cannot provide a specific final disease diagnosis.

4	Anitha Ponraj, Palanigurupackiam Nagaraj, et al.	2024	A multi-patch-based deep learning model with VGG19 for breast cancer classifications in the pathology images	Applies a multi-patch autoencoder combined with VGG19 to capture local cell patterns. **Dataset used:** CBIS-DDSM & MIAS: Public mammography datasets used for multi-patch evaluation.	Evaluation is limited to specific datasets. Performance drops across different microscope magnifications.
5	Istiak Ahmad, Fahad Alqurashi	2024	Early cancer detection using deep learning and medical imaging: A survey	Surveys 99 papers across 12 cancer types using transfer learning and segmentation. **Dataset used:** Various: Survey paper reviewing 99 articles; covers 12 types of cancer across multiple public datasets.	Most reviewed models are specialized for only one cancer type. They lack clinical explainability.
6	Aadhi Aadhavan Balasubramanian, Salah Al-Heejawi, et al.	2024	Ensemble Deep Learning-Based Image Classification for Breast Cancer Subtype and Invasiveness Diagnosis	Combines multiple models (VGG16 and ResNet50) into an ensemble team for better predictions. **Dataset used:** BACH & BreakHis: Used for classifying both invasiveness and cancer subtypes.	Achieves high lab accuracy. Lacks practical software integration into daily hospital workflows.
7	Ardavan Modarres, Erfan Ebrahim Esfahani, Mahsa Bahrami	2023	GROUP MIXER: Patch-Based Group Convolutional Neural Network for Breast Cancer Detection	Builds a lightweight CNN using group convolutions and channel shuffling to save memory. **Dataset used:** BreakHis, spanning from 40x to 400x.	Compressing the model causes a minor drop in overall detection accuracy.

8	Sergio Ortiz, Ignacio Rojas-Valenzuela, Fernando Rojas, et al.	2023	Novel methodology for detecting and localizing cancer area in histopathological images based on overlapping patches	Implements overlapping patch selection with a 3x3 error correction filter. **Dataset used:** HUP & CINIJ which are Datasets from Hospital University of Pennsylvania and National Cancer Institute of Peru.	Results are highly sensitive to threshold tuning. Evaluation is limited to two datasets
9	Aashita Rajput, Muskan Yadav, Sachin Yadav, et al.	2023	Patch-Based Breast Cancer Histopathological Image Classification using Deep Learning	Uses standard CNN architectures to classify biopsy images into healthy or cancerous regions. **Dataset used:** BreakHis & BUSI which are used for microscopic biopsy and ultrasound image classification.	The architecture is too simple. It cannot handle complex biological variations in tissue.
10	Yinsheng He, Xingyu Li	2023	Whole-slide-imaging Cancer Metastases Detection and Localization with Limited Tumorous Data	Utilizes reverse knowledge distillation to train models with 90% less labeled tumor data. **Dataset used:** Camelyon16 & WSSS4LUAD which focused on detecting metastases with very few labeled samples.	Requires a small amount of tumor data to start. Does not significantly outperform larger models.

11	Ahad Alloqmani, Yoosef B. Abushark, Asif Irshad Khan	2023	Anomaly Detection of Breast Cancer Using Deep Learning	Employs MobileNetV2 to capture normal features and flags deviations as anomalies. **Dataset used**: INbreast & MIAS are Mammography datasets used to detect abnormalities from "normal" data.	Performance relies heavily on public datasets. These may not match messy real-world clinical images.
12	Mai Alsammak, Marwa Khattabi	2023	Breast cancer screening, a literature review	Evaluates global screening data from traditional tools like mammography and MRI. **Dataset used:** GLOBOCAN Database- Review focused on global incidence and mortality statistics.	Manual screening has high false-positive rates. Low-income areas lack automated AI screening access.
13	Xiaoyan Jiang, Zuojin Hu, Shuihua Wang, Yudong Zhang	2023	Deep Learning for Medical Image-Based Cancer Diagnosis	Reviews clinical applications of major architectures like CNNs and GANs across multiple scans. **Dataset used:** Various datasets, Review paper summarizing common radiological and histopathology data sources.	Deep architectures often overfit training data. Models act as untrustworthy, unexplainable "black boxes."

14	Hirald Dwaraka Praveena, Shaik Saba Samreen, et al.	2023	Detection of Breast Cancer Using Deep Learning Techniques	Uses VGG16 and standard CNN layers to isolate invasive ductal carcinoma (IDC) regions. **Dataset used:** Breast Histopathology Images which used Large-scale dataset of 277,524 patches for IDC classification.	Classification accuracy fluctuates significantly depending on the microscope magnification level.
15	Salman Zakareya, Habib Izadkhah, Jaber Karimpour	2023	A New Deep-Learning-Based Model for Breast Cancer Diagnosis from Medical Images	Develops a hybrid network using granular computing and visual attention blocks. **Dataset used:** BreakHis (Histopathology) & BUSI (Ultrasound), Hybrid evaluation using two modalities.	Model performance is heavily limited by the scarcity of high-quality, annotated medical training data.
16	Monika Subramanian, Nagarajan Ganesan, SathishKumar Balasubramaniyan	2023	An Extensive Survey on Various Tumor Detection in Histopathological Images Using Deep Learning Techniques	Evaluates high-resolution slide filtering and segmentation techniques using convolutional features. **Dataset used:** Various: Survey focused on Whole Slide Imaging (WSI) and high-resolution histology data.	Extracting deep features from gigapixel whole-slide images is computationally slow and expensive.
17	Yawen Wu, Michael Cheng, Shuo Huang, et al.	2022	Recent Advances of Deep Learning for Computational Histopathology:	Surveys deep learning workflows for color normalization, tissue segmentation, and cancer prognosis.	Severe tissue staining variations across different laboratories prevent stable

			Principles and Applications	**Dataset used:** Various: Review paper focusing on H&E-stained whole slide images (WSIs).	global model generalization.
18	Alhassan Ali Ahmed, Mohamed Abouzid, Elżbieta Kaczmarek	2022	Deep Learning Approaches in Histopathology	Explores the clinical role of deep learning in automated tumor grading and slide evaluation. **Dataset used:** Various: Review paper covering major public datasets like TCGA and Camelyon.	Models face strict clinical validation hurdles. Pathologists remain hesitant to accept AI results.
19	Murtaza Ashraf, Willmer Rafell Quiñones Robles, Mujin Kim, Young Sin Ko, Mun Yong Yi	2022	A loss-based patch label denoising method for improving whole-slide image analysis using a convolutional neural network	Proposes "LossDiff" to automatically detect and discard mislabeled training patches. **Dataset used:** Private Dataset of 206 stomach cancer WSIs; also validated on Camelyon16.	The noise-filtering threshold must be adjusted manually. This risks dropping rare cell details.
20	Ozan Ciga, Tony Xu, Sharon Nofech-Mozes, Shawna Noy, Fang-I Lu, Anne L. Martel	2021	Overcoming the limitations of patch-based learning to detect cancer in whole slide images	Uses a new patch-selection strategy to train networks with lower false-positive rates. **Dataset used:** Private Clinical Dataset consisting Post-neoadjuvant chemotherapy WSIs; also used Camelyon16/17.	Performs well at patch levels but struggles to map out the entire tumor bed boundaries accurately.
21	Massimo Salvi, Filippo Molinari, U	2021	Impact of stain normalization and patch selection on	Implements color stain normalization and	Performance depends entirely on the accuracy of the

	Rajendra Acharya, Luca Molinaro, Kristen M. Meiburger		the performance of convolutional neural networks	filters patches based on cell nuclei count. **Dataset used:** TCGA (The Cancer Genome Atlas) consisting WSIs for both breast and prostate cancer.	initial automated nuclear segmentation step.
22	Irum Hirra, Mubashir Ahmad, Ayaz Hussain, M. Usman Ashraf, Iftikhar Ahmed Saeed, Syed Furqan Qadri, Ahmed M. Alghamdi	2021	Breast Cancer Classification From Histopathological Images Using Patch-Based Deep Learning Modeling	Builds an unsupervised Deep Belief Network to learn local features from patches. **Dataset used:** BreakHis which was Evaluated using multiple magnification levels (40x, 100x, 200x, 400x).	Requires heavy GPU computing power. Cannot determine specific or complex tumor subtypes.
23	Yasin Yari, Thuy V. Nguyen, Hieu T. Nguyen	2020	Deep Learning Applied for Histological Diagnosis of Breast Cancer	Trains standard CNN models on tissue shapes to automate routine slide scanning. **Dataset used:** ICIAR 2015 consisting 249 images across four breast cancer histology categories.	Severely limited by the global shortage of publicly accessible, high-quality histopathology samples.
24	Yuqian Li, Junmin Wu, Qisong Wu	2019	Classification of Breast Cancer Histology Images Using Multi-Size and Discriminative Patches Based on Deep Learning	Extracts multi-size patches simultaneously to evaluate cellular and larger tissue context together. **Dataset used:** ICIAR 2018 (BACH) consisting 400 images	Handling multiple image resolutions at once complicates the data preprocessing pipeline.

				(Normal, Benign, In Situ, Invasive).	
25	Md Zahangir Alom, Chris Yakopcic, et al.	2019	Breast Cancer Classification from Histopathological Images with Inception Recurrent Residual Convolutional Neural Network	Merges Inception-v4, ResNet, and Recurrent CNN configurations into a massive hybrid architecture. **Datasets used:** BreakHis & ICIAR 2015: Combined two major public challenges for benchmarking.	The highly complex model requires substantial datasets and long hours to train properly.

2.3 LITERATURE REVIEW SUMMARY

The literature review highlights significant advancements in the application of deep learning techniques for breast cancer detection using histopathological and medical imaging data. Various studies have demonstrated the effectiveness of Convolutional Neural Networks (CNNs), transfer learning, and ensemble learning models in improving classification accuracy and diagnostic performance. Advanced architectures such as ResNet, DenseNet, EfficientNet, VGG19, and hybrid CNN models have been widely explored for efficient feature extraction and tissue classification. Patch-based learning approaches have gained importance for processing large Whole Slide Images (WSIs) by enabling localized analysis of tissue regions. Several researchers have also incorporated preprocessing methods such as stain normalization, noise reduction, and data augmentation to improve model robustness and generalization capability. In addition, anomaly detection and weakly supervised learning techniques have been introduced to reduce dependency on detailed annotations and improve scalability in medical image analysis. Explainable AI techniques such as Grad-CAM have also been explored to improve transparency and interpretability in AI-based diagnostic systems. However, many existing systems still depend heavily on large labeled datasets and struggle to generalize across different imaging conditions and datasets.Furthermore, issues related to high computational complexity, limited interpretability, and difficulty in detecting rare or unseen cancer patterns remain major challenges. These research gaps highlight the need for a more robust, scalable, and interpretable framework for efficient breast cancer detection.

2.4 GAPS IDENTIFIED

1. Most existing models need a lot of detailed labeled data, which takes a lot of time and expert effort.
2. Many current methods cannot detect new or rare cancer patterns because they only learn from given data.
3. Patch-based models often miss the overall tissue context, which can lead to wrong predictions.
4. Many models do not work well on different datasets and also do not clearly explain their predictions.

2.5 EXISTING SYSTEM

Several AI-based platforms have been developed to support breast cancer detection and histopathological image analysis using deep learning techniques. Paige Breast uses cloud-based AI models to detect metastases and analyze tissue abnormalities across large and diverse datasets. Ibex Medical Analytics (Galen Breast) applies CNN-based architectures trained on extensive annotated datasets to identify multiple breast tissue features and assist pathologists in diagnosis. Aiforia Breast Cancer Suite automates cancer grading and tissue analysis using supervised deep learning models that evaluate cellular characteristics and morphological patterns. PathAI (AISight) employs ensemble CNN models for rapid case prioritization, cancer detection, and cell identification across varied patient populations. Cytomine, an open-source collaborative platform, supports large-scale tissue segmentation and integration of custom deep learning models for medical image analysis. These systems improve diagnostic efficiency and reduce manual workload for pathologists. However, many existing platforms still face challenges related to dataset dependency, interpretability, computational complexity, and generalization across different clinical environments.

2.6 PROBLEM STATEMENT

Existing deep learning models for histopathological image analysis face several limitations in real-world clinical applications. Most systems depend on large amounts of labeled data, making the annotation process time-consuming and expensive. Many patch-based models fail to capture overall tissue context, which can reduce classification accuracy. In addition, several models generalize poorly across different datasets and imaging conditions. Another major challenge is the lack of interpretability, as most models do not clearly explain their predictions, reducing clinician trust. Therefore, there is a need for a robust and efficient breast cancer detection system that

reduces labeling dependency, improves contextual understanding and generalization, supports anomaly detection, and provides explainable outputs suitable for clinical use.

2.7 PROPOSED SYSTEM

The proposed system is a patch-based cancer detection framework designed to analyze Whole Slide Images (WSIs) using deep learning. The system begins by extracting meaningful patches from WSIs and applying preprocessing techniques such as normalization, noise reduction, and data augmentation to improve data quality. These patches are then processed using advanced CNN models like ResNet, DenseNet, and EfficientNet with transfer learning for effective feature extraction and classification of tumor and normal tissue. A weakly supervised approach is adopted using slide-level labels to reduce annotation dependency. The system also incorporates anomaly detection to identify abnormal tissue patterns and assess risk levels. Explainable AI using Grad-CAM is integrated to generate heatmaps for better interpretability and clinical trust. Finally, the models are evaluated using standard performance metrics to identify the most efficient architecture, and the system provides predictions along with visual explanations for practical use.

2.8 OBJECTIVES

1. To design and implement a patch-based system for cancer detection by extracting meaningful patches from Whole Slide Images (WSI) and applying essential preprocessing techniques.
2. To develop an EfficientNet-based deep learning model using transfer learning for tumor detection from image patches, and to perform anomaly analysis and risk prediction based on learned tissue patterns.
3. To evaluate and compare the performance of deep learning models such as ResNet, DenseNet, and EfficientNet using standard performance metrics, while integrating Explainable AI (Grad-CAM) techniques to improve model transparency.
4. To develop a web application for deploying the trained model and displaying prediction results in a user-friendly manner.

CHAPTER 3

SYSTEM REQUIREMENTS AND SPECIFICATION

3.1 HARDWARE REQUIREMENTS

- Processor: Intel Core i7 / AMD Ryzen 7 (8+ cores recommended).
- RAM: 16GB minimum (32GB recommended for high-resolution slides).
- Storage: 1TB+ HDD/SSD for massive WSI datasets.
- GPU: NVIDIA GPU with 8GB+ VRAM (e.g., RTX 3060).

3.2 SOFTWARE REQUIREMENTS

- Operating System: Windows 10/11
- Programming Language: Python 3.9+
- Deep Learning Frameworks: TensorFlow
- WSI Processing Library: OpenSlide Library (.svs, .tif support)
- Image Processing: OpenCV and Albumentations
- Explainable AI (XAI): Grad-CAM
- Medical Visualization: QuPath
- Web Application Framework: Streamlit
- IDE: VS Code

CHAPTER 4

METHODOLOGY

4.1 METHODOLOGY DIAGRAM

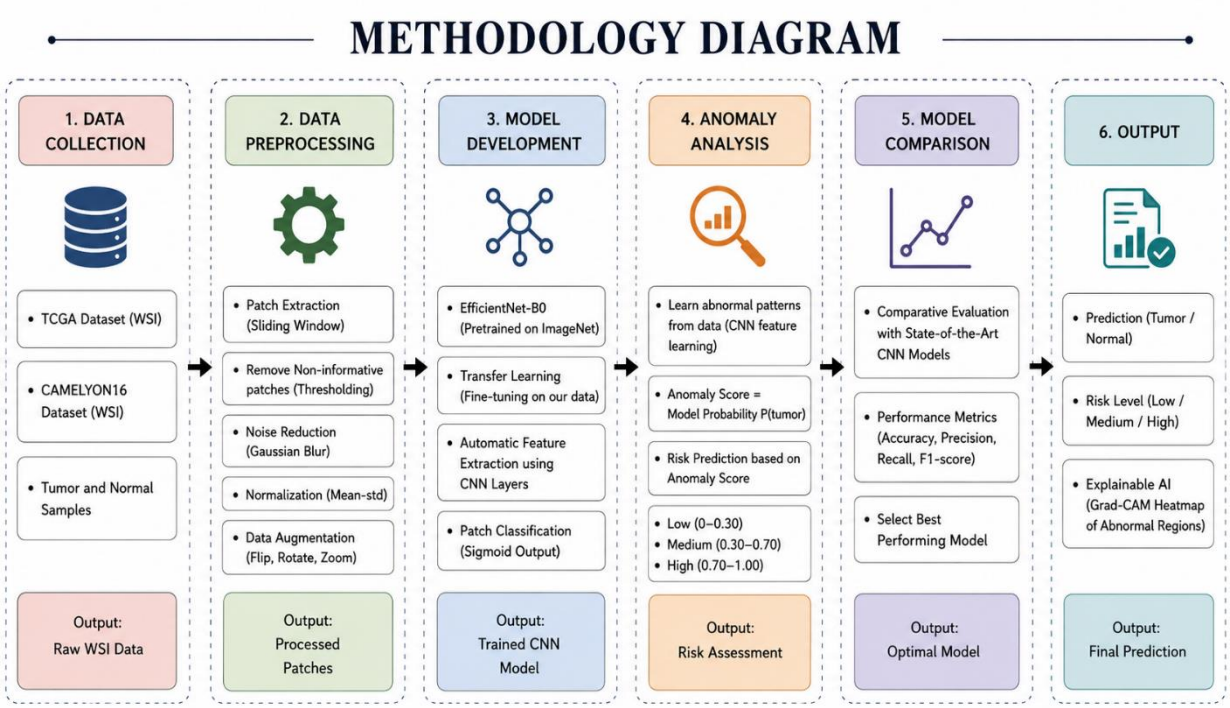


Fig 4.1: Methodology diagram

4.2 DETAILED METHODOLOGY

The proposed methodology presents a patch-based unimodal breast cancer detection framework using Whole Slide Histopathological Images (WSIs) and deep learning techniques. The system is designed to identify abnormal tissue regions by combining image preprocessing, patch extraction, transfer learning, anomaly analysis, model comparison, and Explainable Artificial Intelligence (XAI). Initially, high-resolution histopathological datasets such as TCGA and CAMELYON16 are collected and preprocessed using techniques including patch extraction, thresholding, Gaussian

blur filtering, normalization, and data augmentation to improve image quality and model generalization.

The extracted image patches are then used to train deep learning architectures such as EfficientNet-B0, ResNet, and DenseNet using transfer learning. EfficientNet-B0 is used as the primary model for feature extraction and classification of tumor and normal tissue patches. The framework further incorporates anomaly detection and risk analysis to identify abnormal tissue patterns and classify them into different risk levels. Grad-CAM based Explainable AI techniques are integrated to generate heatmaps highlighting important tissue regions influencing model predictions. Finally, the performance of different models is compared using standard performance metrics to identify the most effective architecture, and the final predictions are deployed through a web application or API for practical clinical usage.

Step 1 — Data Collection

The system takes histopathological Whole Slide Images (WSIs) as input from publicly available datasets such as TCGA and CAMELYON16. These datasets contain high-resolution tissue images collected under different imaging conditions and magnification levels.

Both tumor and normal tissue samples are included to ensure balanced learning during model training. The datasets are stored in high-resolution formats such as .svs and .tif, which preserve important cellular and structural details required for accurate cancer detection.

Dataset diversity is maintained by including images from different patients, scanners, and staining conditions. This helps improve model generalization and robustness across different clinical scenarios. The collected datasets are organized and labeled appropriately before preprocessing and model training.

Step 2 — Data Preprocessing

The preprocessing module is developed to standardize and prepare the histopathological images before training. Since Whole Slide Images are extremely large, each image is divided into smaller 224 × 224 image patches using a sliding window approach. This patch extraction process reduces computational complexity and enables efficient processing of high-resolution tissue images.

Background or empty patches containing very little tissue information are removed using thresholding techniques. Gaussian Blur filtering is applied to reduce image noise and smooth unwanted variations in tissue appearance.

Normalization is then performed using Mean–Standard Deviation normalization to standardize pixel intensity distributions across all images. Since histopathological images often contain staining and illumination variations, normalization helps maintain consistency during training.

Data augmentation techniques such as horizontal flipping, rotation, zooming, and random cropping are applied only to the training data. These operations increase dataset diversity, reduce overfitting, and improve model generalization capability. After preprocessing, the extracted image patches are used as input for deep learning model training.

Step 3 — Model Development

The model development stage utilizes EfficientNet-B0 as the primary deep learning architecture. EfficientNet-B0 is initialized using ImageNet pre-trained weights and fine-tuned using transfer learning on the histopathological image dataset.

The CNN layers automatically learn important pathological features such as nuclei arrangement, tissue texture, cellular morphology, and abnormal tissue patterns from the extracted image patches. Transfer learning significantly reduces training time and improves classification performance even with limited medical datasets.

The final classification layer uses a sigmoid activation function to classify image patches into two categories:

- Tumor
- Normal

In addition to EfficientNet-B0, other architectures such as ResNet and DenseNet are also implemented for comparative analysis and performance evaluation. These models are trained and evaluated under similar conditions to identify the most effective architecture for breast cancer detection.

Step 4 — Anomaly and Risk Analysis

The proposed system integrates anomaly detection techniques to identify abnormal tissue patterns in histopathological images. The trained model learns important tissue characteristics and detects deviations as potential tumor regions.

An anomaly score is generated based on the probability of tumor prediction:

P(tumor)P(\text{tumor})P(tumor)

Based on the anomaly score, the detected tissue regions are categorized into three risk levels:

- Low Risk
- Medium Risk
- High Risk

This risk-level prediction improves clinical interpretation and assists pathologists in identifying suspicious tissue regions more efficiently. The anomaly analysis module helps reduce false predictions and improves diagnostic reliability.

Step 5 — Model Comparison

Multiple deep learning architectures including EfficientNet-B0, ResNet, and DenseNet are compared to identify the most effective model for breast cancer detection.

The performance of each model is evaluated using standard performance metrics to measure classification efficiency, robustness, and prediction reliability. Comparative analysis is performed to identify the architecture that provides the best balance between classification accuracy, computational efficiency, and generalization capability.

The best-performing model is selected for final deployment and prediction generation.

Step 6 — Output Generation

The final output module generates classification results for the input histopathological images. Each image patch is classified as:

- Tumor
- Normal

The system also generates:

- Risk-level prediction (Low / Medium / High)
- Grad-CAM heatmaps for visual interpretability
- Highlighted tissue regions influencing model decisions

Grad-CAM is used to visualize the important pathological regions responsible for model predictions, thereby improving transparency and trustworthiness in clinical applications.

Finally, the generated results can be deployed through a web application or API interface to enable user interaction and assist pathologists in practical diagnostic workflows.

CONCLUSION

The project “Patch-Based Unimodal Breast Cancer Detection from Whole Slide Images Using CNN” has successfully completed its initial phase, which included an extensive literature review, analysis of existing breast cancer detection techniques, identification of major research gaps, and formulation of a clear problem statement along with specific project objectives. Through the review of various deep learning, patch-based learning, anomaly detection, and histopathological image analysis methods, the study identified challenges such as high computational complexity of Whole Slide Images, dependency on large annotated datasets, lack of interpretability in deep learning models, and limitations in capturing contextual tissue information. Based on these observations, a robust conceptual framework for the proposed system has been established. The methodology involving image preprocessing, patch extraction, transfer learning using CNN architectures, anomaly detection, risk analysis, and Explainable Artificial Intelligence techniques such as Grad-CAM has been finalized for implementation. The initial phase has therefore laid a strong theoretical and technical foundation for building an interpretable, scalable, and clinically supportive breast cancer detection system; however, the actual implementation, training, and performance evaluation of the system have not yet been carried out and remain as part of the next phase of the work.

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